

Subqueries In SQL Assignment

Employee Dataset:

emp_id	name	department_id	salary
101	Abhishek	D01	62000
102	Shubham	D01	58000
103	Priya	D02	67000
104	Rohit	D02	64000
105	Neha	D03	72000
106	Aman	D03	55000
107	Ravi	D04	60000
108	Sneha	D04	75000
109	Kiran	D05	70000
110	Tanuja	D05	65000

Department Dataset:

department_id	department_name	location
D01	Sales	Mumbai
D02	Marketing	Delhi
D03	Finance	Pune
D04	HR	Bengaluru
D05	IT	Hyderabad

Sales Dataset:

sale_id	emp_id	sale_amount	sale_date
201	101	4500	2025-01-05
202	102	7800	2025-01-10
203	103	6700	2025-01-14
204	104	12000	2025-01-20
205	105	9800	2025-02-02
206	106	10500	2025-02-05
207	107	3200	2025-02-09
208	108	5100	2025-02-15
209	109	3900	2025-02-20
210	110	7200	2025-03-01

Basic Level

1. Retrieve the names of employees who earn more than the average salary of all employees.

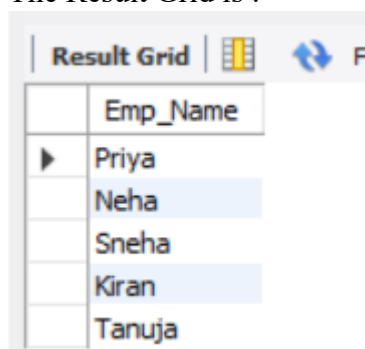
Answer: The SQL Query we use to resolve the given situation is:

```
Select Emp_Name
```

```
From employee_dataset
```

```
Where Emp_Salary > (Select Avg(Emp_Salary) from employee_dataset);
```

The Result Grid is :



	Emp_Name
▶	Priya
	Neha
	Sneha
	Kiran
	Tanuja

2. Find the employees who belong to the department with the highest average salary.

Answer: The SQL Query we use to resolve the given situation is:

```
SELECT e.emp_name, d.department_name, e.Emp_salary
```

```
FROM employee_dataset e
```

```
JOIN department_dataset d
```

```
ON e.department_id = d.department_id
```

```
WHERE e.department_id IN (
```

```
SELECT department_id
```

```
FROM employee_dataset
```

```
GROUP BY department_id
```

```
HAVING AVG(emp_salary) = (
```

```
SELECT MAX(avg_sal)
```

```
FROM (
```

```
SELECT AVG(emp_salary) AS avg_sal
```

```
FROM employee_dataset
```

```

GROUP BY department_id
) AS dept_avgs
)
);

```

The Result Grid is:

Result Grid   Filter Rows:			
	emp_name	department_name	Emp_salary
▶	Ravi	HR	60000
	Sneha	HR	75000
	Kiran	IT	70000
	Tanuja	IT	65000

3. List all employees who have made at least one sale.

Answer: The SQL Query we use to resolve the given situation is:

```

Select distinct e.emp_id, e.Emp_name
from employee_dataset e
Join Sales_department s
on e.emp_id = s.emp_id;

```

The Result Grid is:

Result Grid   Filter		
	emp_id	Emp_name
▶	101	Abhishek
	102	Shubham
	103	Priya
	104	Rohit
	105	Neha
	106	Aman
	107	Ravi
	108	Sneha
	109	Kiran
	110	Tanuja

4. Find the employee with the highest sale amount.

Answer: The SQL Query we use to resolve the given situation is:

```
SELECT e.emp_id, e.emp_name, s.sale_amount
FROM employee_dataset e
JOIN sales_department s
    ON e.emp_id = s.emp_id
WHERE s.sale_amount = (
    SELECT MAX(sale_amount) FROM sales_department
);
```

The Result Grid is:

Result Grid			
	emp_id	emp_name	sale_amount
▶	104	Rohit	12000

5. Retrieve the names of employees whose salaries are higher than Shubham's salary.

Answer: The SQL Query we use to resolve the given situation is:

```
SELECT emp_name, emp_salary
FROM employee_dataset
WHERE emp_salary > (
    SELECT emp_salary
    FROM employee_dataset
    WHERE emp_name = 'Shubham'
);
```

The Result Grid is:

Result Grid		
	emp_name	emp_salary
▶	Abhishek	62000
	Priya	67000
	Rohit	64000
	Neha	72000
	Ravi	60000
	Sneha	75000
	Kiran	70000
	Tanuja	65000

Intermediate Level

1. Find employees who work in the same department as Abhishek.

Answer: The SQL Query we use to resolve the given situation is:

```
SELECT emp_name, department_id
FROM employee_dataset
WHERE department_id = (
    SELECT department_id
    FROM employee_dataset
    WHERE emp_name = 'Abhishek'
)
AND emp_name != 'Abhishek';
```

Result Grid:

Result Grid			Filter Rows:
	emp_name	department_id	
▶	Shubham	D01	

2. List departments that have at least one employee earning more than ₹60,000.

Answer: The SQL Query we use to resolve the given situation is:

```
SELECT department_name
FROM department_dataset
WHERE department_id IN (
    SELECT department_id
    FROM employee_dataset
    WHERE emp_salary > 60000
);
```

Result Grid:

	department_name
▶	Sales
	Marketing
	Finance
	HR
	IT

3. Find the department name of the employee who made the highest sale.

Answer: The SQL Query we use to resolve the given situation is:

```
SELECT d.department_name
FROM department_dataset d
JOIN employee_dataset e
    ON d.department_id = e.department_id
JOIN sales_department s
    ON e.emp_id = s.emp_id
WHERE s.sale_amount = (
    SELECT MAX(sale_amount) FROM sales_department
);
```

Result Grid:

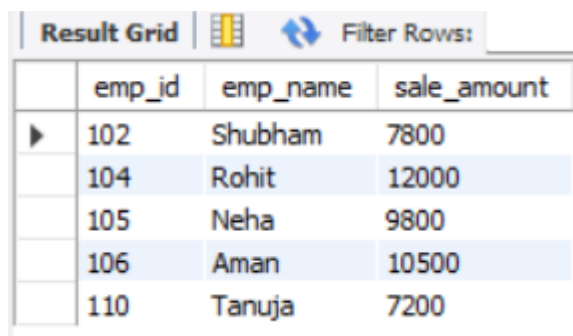
	department_name
▶	Marketing

4. Retrieve employees who have made sales greater than the average sale amount.

Answer: The SQL Query we use to resolve the given situation is:

```
SELECT e.emp_id, e.emp_name, s.sale_amount
FROM employee_dataset e
JOIN sales_Department s
  ON e.emp_id = s.emp_id
WHERE s.sale_amount > (
  SELECT AVG(sale_amount) FROM sales_Department
);
```

Result Grid:



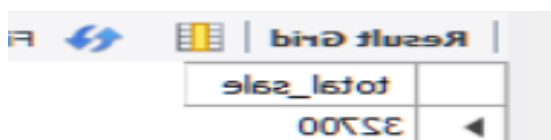
The screenshot shows a 'Result Grid' window with a toolbar containing icons for 'Filter Rows' and 'Refresh'. The grid displays the following data:

	emp_id	emp_name	sale_amount
▶	102	Shubham	7800
	104	Rohit	12000
	105	Neha	9800
	106	Aman	10500
	110	Tanuja	7200

5. Find the total sales made by employees who earn more than the average salary.

Answer: The SQL Query we use to resolve the given situation is:

```
SELECT SUM(s.sale_amount) AS total_sale
FROM sales_department s
WHERE s.Emp_id IN (
  SELECT Emp_id
  FROM employee_dataset
  WHERE Emp_salary > (SELECT AVG(Emp_salary) FROM employee_dataset)
);
```



The screenshot shows a 'Result Grid' window with a toolbar containing icons for 'Filter Rows', 'Refresh', and 'Grid'. The grid displays the following data:

	total_sale
◀	32700

Result Grid:

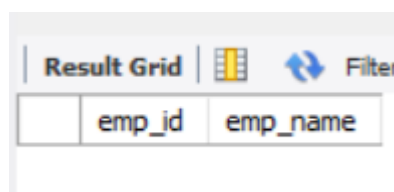
Advanced Level

1. Find employees who have not made any sales.

Answer: The SQL Query we use to resolve the given situation is:

```
SELECT emp_id, emp_name
FROM employee_dataset
WHERE emp_id NOT IN (
    SELECT emp_id
    FROM sales_department
);
```

Result Grid:



emp_id	emp_name
--------	----------

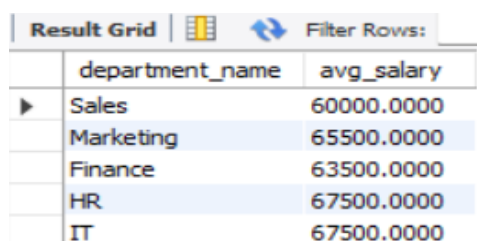
(As every Emp have made a sale hence the result is empty)

2. List departments where the average salary is above ₹55,000.

Answer: The SQL Query we use to resolve the given situation is:

```
SELECT d.department_name, AVG(e.emp_salary) AS avg_salary
FROM employee_dataset e
JOIN department_dataset d
    ON e.department_id = d.department_id
GROUP BY d.department_name
HAVING AVG(e.emp_salary) > 55000;
```

Result Grid:



department_name	avg_salary
Sales	60000.0000
Marketing	65500.0000
Finance	63500.0000
HR	67500.0000
IT	67500.0000

3. Retrieve department names where the total sales exceed ₹10,000.

Answer: The SQL Query we use to resolve the given situation is:

```
SELECT d.department_name, SUM(s.sale_amount) AS total_sales
FROM department_dataset d
JOIN employee_dataset e
    ON d.department_id = e.department_id
JOIN sales_department s
    ON e.emp_id = s.emp_id
GROUP BY d.department_name
HAVING SUM(s.sale_amount) > 10000;
```

Result Grid:

	department_name	total_sales
▶	Sales	12300
	Marketing	18700
	Finance	20300
	IT	11100

4. Find the employee who has made the second-highest sale.

Answer: The SQL Query we use to resolve the given situation is:

```
SELECT e.emp_name, s.sale_amount
FROM employee_dataset e
JOIN sales_department s
    ON e.emp_id = s.emp_id
ORDER BY s.sale_amount DESC
LIMIT 1 OFFSET 1;
```

Result Grid:

	emp_name	sale_amount
▶	Aman	10500

5. Retrieve the names of employees whose salary is greater than the highest sale amount recorded.

Answer: The SQL Query we use to resolve the given situation is:

```
SELECT emp_name, emp_salary
```

```
FROM employee_dataset
```

```
WHERE emp_salary > (
```

```
    SELECT MAX(sale_amount)
```

```
    FROM sales_department
```

```
);
```

Result Grid:

Result Grid   Filter Row		
	emp_name	emp_salary
▶	Abhishek	62000
	Shubham	58000
	Priya	67000
	Rohit	64000
	Neha	72000
	Aman	55000
	Ravi	60000
	Sneha	75000
	Kiran	70000
	Tanuja	65000